

TẬP I

Session 21

Instructor: Hồ Lê Vũ



Start recording...

OVERVIEW

- TIME TRAVEL

LEAD-IN

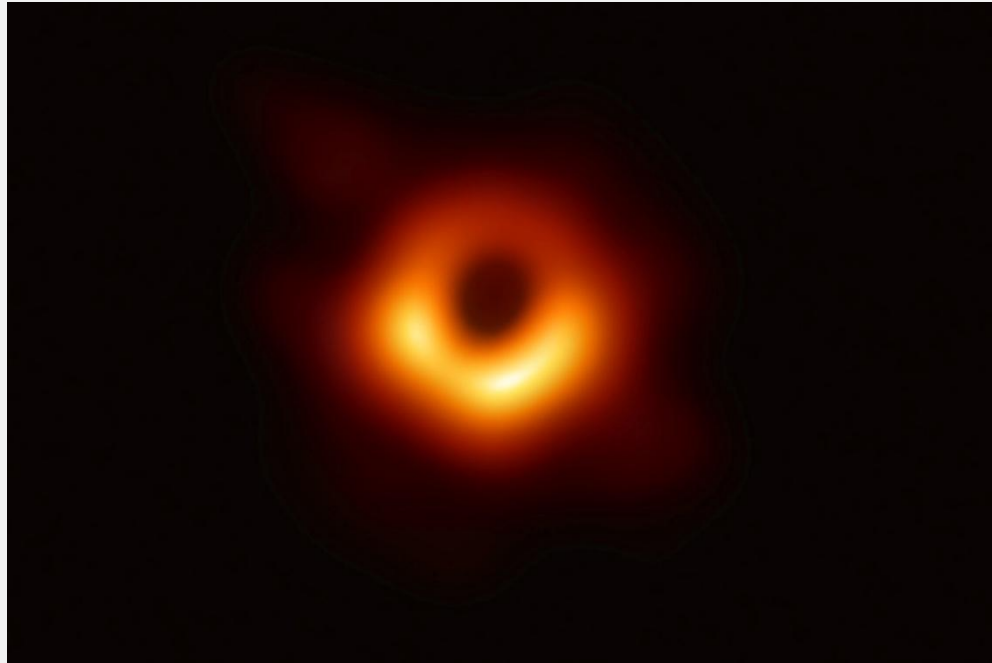
→ *Scientists* and *the time machine*...

- Part 2 Time travel to the future

Content-based approach to learning

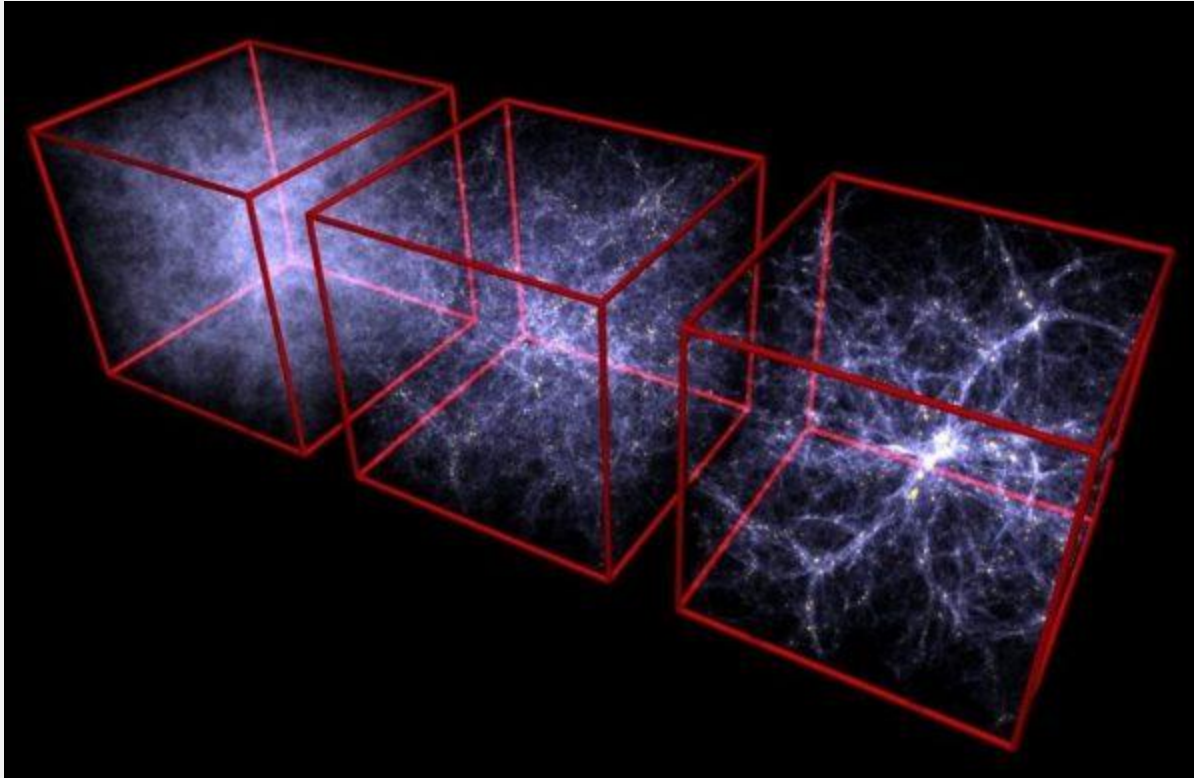
ANNOTATION

- *Time travel & supermassive black hole*



ANNOTATION

- *More information about black hole and dark energy*



DARK ENERGY

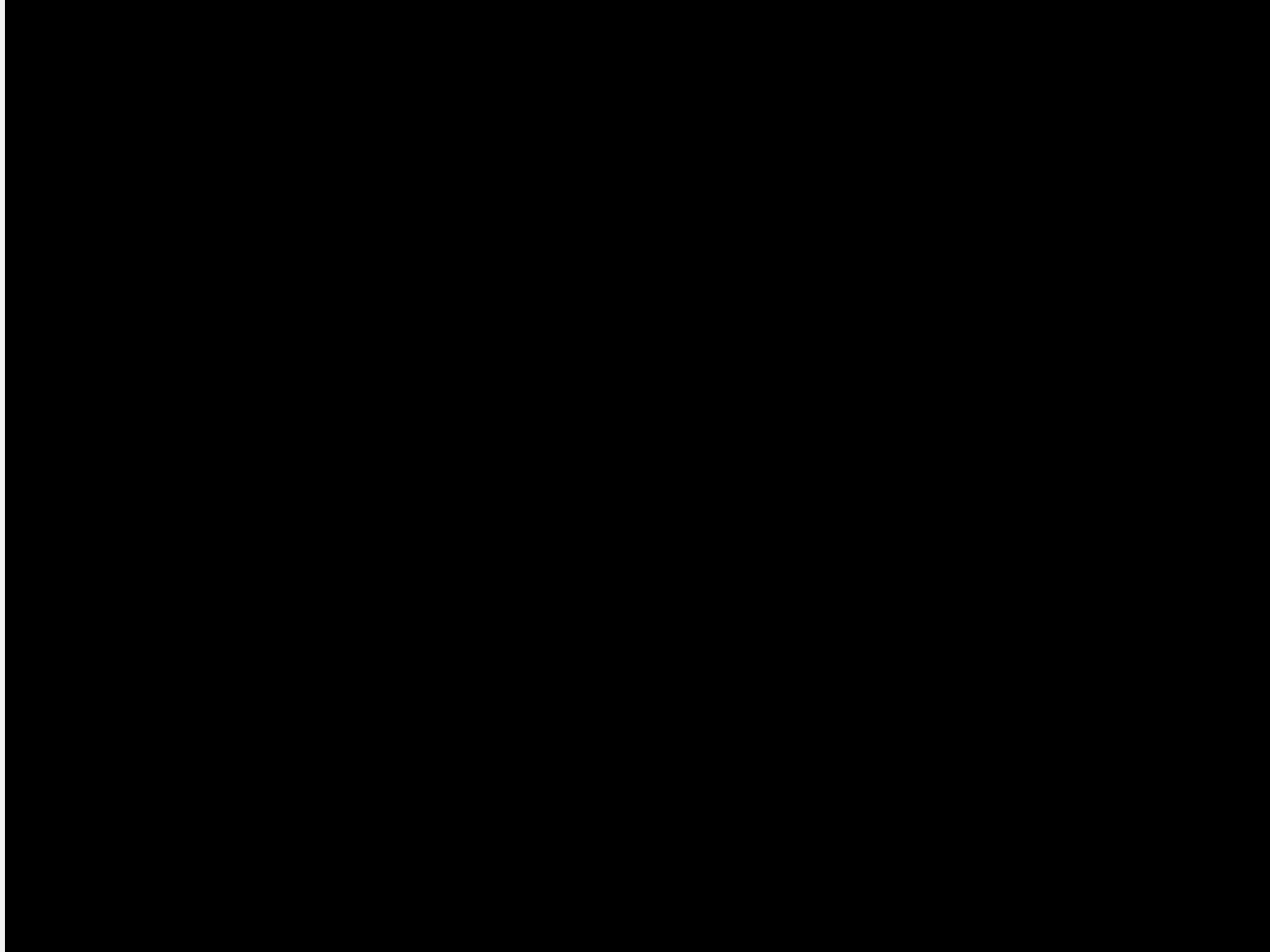
After the Big Bang, the universe began expanding outward. Despite gravity, the force which draws everything together, the expansion of the universe isn't slowing down, it's actually speeding up. Something must be counteracting gravity, a.k.a., dark energy, which could be deemed a property of space. According to Einstein, space is not simply empty and more space could continue to come into existence. Since empty space can have its own energy, as more space emerges, more energy would be added to the universe, increasing its expansion.

DARK MATTER

While dark energy is a force that accounts for the expanding universe, dark matter explains how groups of objects function together. Gravity is expected to cause the centers of galaxies to rotate faster than the outer edges, based on the distribution of the objects inside. Surprisingly, both regions rotated at the same rate, indicating that the spiral galaxies contained additional hidden matter that can't be conventionally observed. Calculating the energy needed to overcome gravity, scientists determined that dark energy makes up roughly 68 percent of the universe. Dark matter makes up another 27 percent, leaving the normal matter that we are familiar with (a.k.a., baryonic matter, composed of protons, neutrons and electrons) to make up less than 5 percent.

Slide No 10

- Extra reading: [A tantalizing 'hint' that astronomers got Dark Energy all wrong](#)



STRUCTURES

- But if we exaggerate it drastically, you can see **the principle at work**.
 - **The principle at work** here was that, within a household, the main income and the social rights were provided by the husband.
- They also give us a **glimpse** of how the universe may end.
 - [literal] He caught a **glimpse** of her in the crowd.
 - [figurative] The film also offers **glimpses** into a celebrity's life.

- The universe would go into reverse and drive everything back to **whence** [O] it came.
 - None of them could answer the basic human questions of **wherefrom** [O] and **whereto** [O]?
- This contract is between Mrs. Green (**hereafter** [F, O, LW] referred to as the vendor) and Mr. Quain (**hereafter** referred to as the purchaser).
- Korea was finally liberated in 1945, and **thereupon** [F, O] the Republic of Korea was established, based upon democratic principles.

FOOD FOR THOUGHT

preposition + *pronoun*: in + this, by + that, from + which

→ *prepositional adverb* + *locative adverb*: in + here, by + there, from + where

→ reversed combination: herein, thereby, wherefrom

→ **pronominal adverb** [F] [L, LW] [O ► archaic]

- Neither party is willing to compromise and **herein** lies the problem.
- Students perform in hospitals, **thereby** gaining a deeper awareness of the therapeutic power of music.
- Several attempts were made to locate the sites, **wherefrom** the earthquake were initiated.

Slide No 14 FOOD FOR THOUGHT

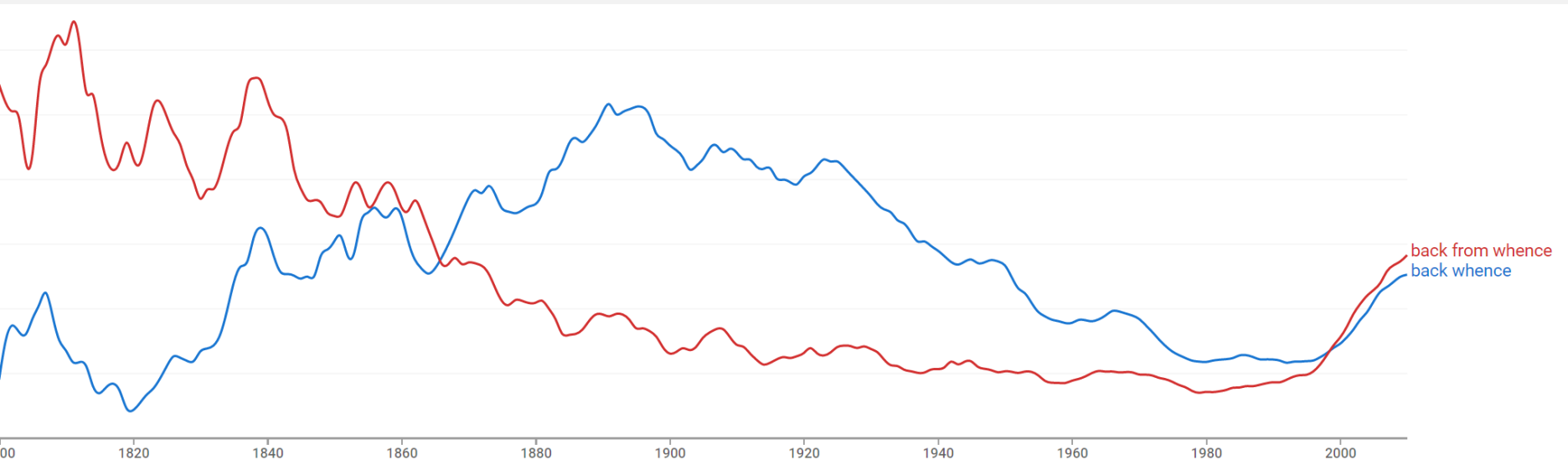
whence = from where [O]

- They send him back **from whence** [O, redundancy] he came.

hence = from here → [physical origin ► logical origin]

- The trade imbalance is likely to rise again. **Hence** [F] a new set of policy actions will be required soon.

thence = from there → [O ► archaic]



ANNOTATION

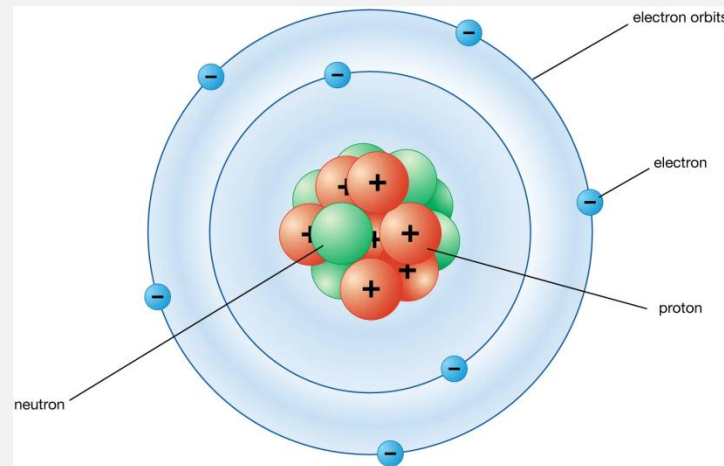
Time travel & the speed of light

LIGHT

At the centre of an **atom** is a **nucleus** containing **protons** and **neutrons**. **Electrons** are arranged around the nucleus in orbits (a.k.a., energy levels). if you energize an atom, you can move its electrons from their natural orbit to higher orbits. Subsequently, whenever an electron in a higher-than-normal orbit falls back to its normal one, it emits a **photon** -- a packet of energy. The photon has a frequency, or color, that exactly matches the distance the electron falls.

Thus, to produce light, one may "excite" the atoms by a flow of electricity to increase their energy. When the energy in the atoms drops (i.e., returning to the original level), they release photons and we have light. A lot of photons give a brighter, more intense type of light. Fewer photons give a dim and less intense light.

With a dual nature as a wave (or electromagnetic radiation) and a particle (or photon), light moves both in a wave-like pattern and as a flow of photons.



HIGGS FIELD

The **Higgs field** is an energy field that exists everywhere in the universe and uses fundamental particles called the **Higgs Boson** to continuously interact with other particles. Every particle moves through this field, interacting with the Higgs Bosons, and as a result, acquires its mass. Different particles interact with the Higgs field with different strengths, which is why some particles are heavier (i.e., have more mass) than others. Photons move through, but **do not interact at all** with the Higgs field (i.e., **have no mass**). This means they aren't bound by any speed limit. They're free to move at the fastest possible speed – the speed of light (186,282 miles/s), a fundamental constant set when our universe was created at the Big Bang.

EINSTEIN'S EQUATION

$E=mc^2$ (E stands for energy, m for mass, & c for the speed of light).

energy: a property of objects, which is transferable via interactions & can be converted into different forms (e.g., heat, light, electrical, kinetic, etc.) but not created or destroyed.

mass: the property of an object commonly taken as a measure of the amount of material it contains and causes it to have weight in a gravitational field.

The speed of light squared is a colossal number, illustrating how much energy there is in even tiny amounts of matter. E.g, 1 gram of water -- if its whole mass were converted into pure energy via $E=mc^2$ -- contains as much energy as 20,000 tons of TNT exploding. That's why such a small amount of uranium or plutonium can produce such a massive atomic explosion. (watch sample clips)

Particles that have mass require energy to accelerate them. The closer to the speed of light you get a particle, the more energy is required to go faster as the particles themselves get more massive (i.e., the faster you go, the heavier you get). That means if you wanted to accelerate a single electron to light speed, you would **need an infinite amount of energy** due to the electron becoming infinitely heavy. **There isn't enough energy in the entire universe to propel just a single electron to the speed of light.**

STRUCTURES

- Believe it or not [I], traveling at near the speed of light transports you to the future.
 - Believe it or not, your brain can see into the future.
 - After losing their first six games, they won the next ten. Go figure [I, AM].

STRUCTURES

- Let's say it gets close, just shy of [I, AM] that ultimate speed.
 - I called him only two minutes shy of midnight.
 - The amount was just shy of \$1 million.
- Everything on the train is in slow motion



STRUCTURES

- After another two years of full thrust, the ship would reach its top speed.
 - All our factories are working at full capacity | at full power.

ANNOTATION

Revisit the fact that it takes light 100,000 years to cross our own Milky Way...

→ *The warp drive as an alternative for space travel*

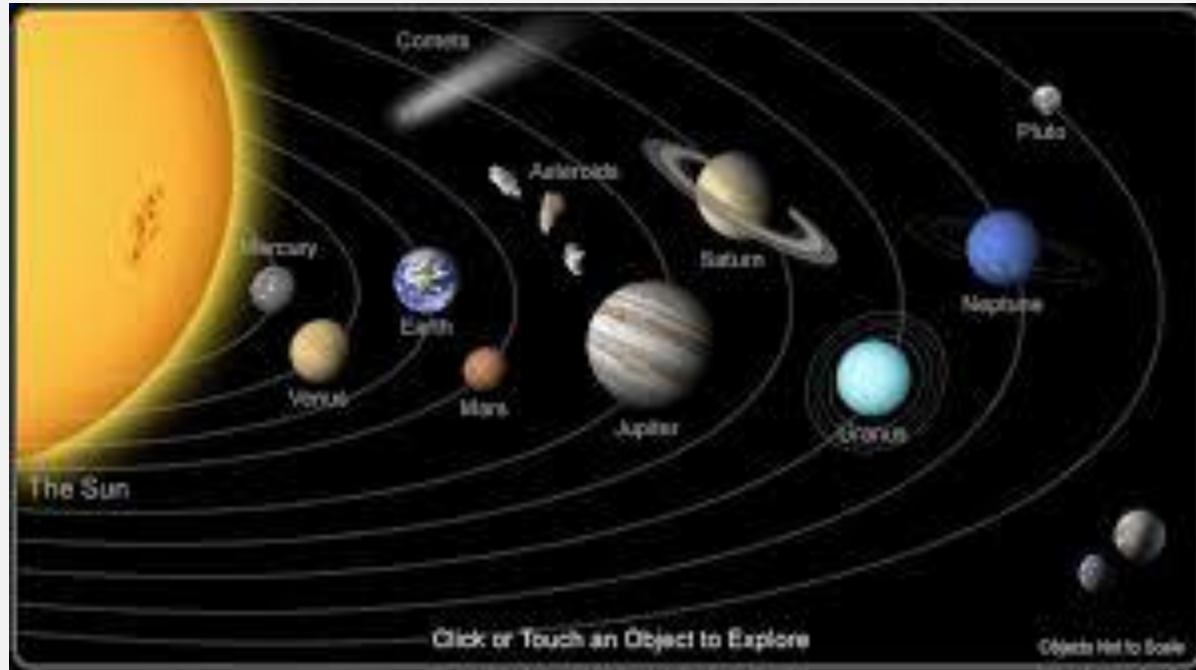
FUN ACTIVITY

Is there a universe creator?

the speed of light (186,282 miles/s), a fundamental constant set when our universe was created at the Big Bang.

ANNOTATION

*Why traveling at nearly the speed of light is key to our **survival***



FUN ACTIVITY

- The concept of **time loop**
→ *watch a romantic comedy about **time loop***

a fictional situation in which a certain length of time (e.g., a few hours or days) repeats over and over. When the time loop "resets", the memories of most characters are reset, and they start afresh. There are usually one or more central characters who retain their memory or become aware of the loop. (*watch clips*)

STRUCTURES

- On second thought(s), I think we should.
 - They're supposed to get married but she's having second thoughts.
- What did you do today? Same old, same old [I, ON].
 - The game's fantasy setting is the same old, same old.
 - It's the same old story [I, ON]: no shortage of candidates, just a shortage of high-quality candidates.

STRUCTURES

- Hang on to him. He's a real **find** [N, functional shift].
 - Another of his lucky **finds** was a pair of candle-holders.
- vs. This disc is a **keeper** [I] and one that belongs on every serious DVD collector's shelf'
 - If he's a good lover and a great listener, he's a **keeper**.
- **find** vs. **keeper**
- *Notes:*
 - If the money remains unclaimed for a certain time it is returned to the **finder** [N].
 - the **keeper** of a lighthouse [N]
 - For a **goalkeeper**, it's a great advantage to have big hands.

WRAP-UP

Brain & time relativity...

STRUCTURES

- I say finders keepers [l, child].
 - Finders keepers, losers weepers [rhyming]
 - Since these islands are not in anyone's jurisdiction [F, LW], it means 'finders keepers' to any existing treasure.

HOMEWORK

- Session 22
 - Khởi Động Bảo Hiểm Thất Nghiệp
 - Final Review
- Session 23
 - SKINCARE